The Macroeconomic Effects of Lifting Barriers to Migration

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Abstract

Many economic studies have identified barriers to migration which can distort labor allocation and productivity, but comparing the macroeconomic importance of each barrier is complicated by variation in study design and sampling. We build and estimate a dynamic general-equilibrium model of rural–urban migration which nests several of the key frictions emphasized in the literature. We discipline the model using large-scale data from Kenya—the site of several recent migration experiments—and simulate each experiment within the model. We proceed to estimate the long-run treatment effects of each intervention, the macroeconomic effects of scaling interventions to the national level, and the aggregate importance of each migration friction. Our findings shed light on the lack of a migration response to large unconditional cash transfers, and reveal how treatment impacts on migration can rise over time as the constraints relaxed by an intervention interact with subsequent productivity shocks.

Keywords: internal migration, urban–rural income gap, general equilibrium model

JEL classification: O15, R23, O18, C63, J61, D58

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1 Introduction

Large gaps in income between rural and urban areas persist in many low-income countries.[1] A growing body of microeconomic evidence points to multiple frictions that can prevent rural households from migrating to cities—monetary costs of migration, information frictions, urban employment risk, and borrowing constraints, for instance—and identifies large monetary returns to relaxing those frictions through targeted interventions (Bryan, Chowdhury, and Mobarak 2014; Akram, Chowdhury, and Mobarak 2017; Baseler 2023; Barnett-Howell et al. 2026; Miner 2026). However, comparing or aggregating results across studies is not straightforward because of differences in sampling and intervention design. Furthermore, we lack a unified quantitative framework to measure the aggregate importance of each barrier to migration, to understand how these frictions interact in general equilibrium, and to evaluate the macroeconomic consequences of scaling up experimental interventions.

We take a step toward filling this gap by building and estimating a quantitative general-equilibrium model of internal migration tailored to randomized controlled trial (RCT) evidence from Kenya. Kenya is the site of several recent experiments that intervene on a broad set of potential migration frictions, and restricting our focus to one country reduces the complexity in modeling sampling variation across studies. The model is designed to nest the main channels emphasized in this experimental work, and to be rich enough to reproduce their interventions within a single structural environment. We estimate the model to match moments from cross-sectional Kenyan data and confront it with experimental evidence from cash transfers, unemployment-insurance, and information interventions. We then use the estimated model to (i) evaluate the long-run effects of each intervention, (ii) evaluate the macroeconomic effects of scaling up the interventions to a national level, and (iii) assess the aggregate importance of different frictions.

Our framework is a dynamic heterogeneous agent model with incomplete markets in the spirit of Aiyagari (1994) with self-selection into location and occupation in the spirit of Roy (1951). A continuum of infinitely lived heterogeneous households chooses consumption, savings, and location between a rural region and an urban region. Borrowing is limited by a borrowing constraint. Urban workers face job-finding and separation shocks, allowing us to model the value of unemployment insurance. Migration to the city imposes a utility cost which is dependent on past urban experi-

1. See Lagakos (2020) for an overview.

ence. Urban experience accumulates stochastically while in the city and can depreciate when returning to the countryside, as in Lagakos, Mobarak, and Waugh (2023). Rural households underestimate their potential urban income, as in Barnett-Howell et al. (2026). Finally, a monetary migration cost picks up all other forces restricting rural–urban migration.

A central design objective of the model is to represent several experiments within a single structural environment. We start by estimating the model using the simulated method of moments, targeting a set of moments from observational data in Kenya. Next, we map the sampling strategy, treatments, and outcome measurement from six existing randomized interventions conducted in Kenya to their approximate analogs in the model. We explicitly reproduce three classes of interventions. First, an unconditional cash transfer, as trialed by Egger et al. (2022) and Miner (2026). Second, an unemployment insurance benefit trialed by Miner (2026). Third, an information treatment that updates rural households' beliefs about urban incomes, as in Barnett-Howell et al. (2026). Crucially, we do not attempt to calibrate the treatment design to match the experimental treatment effects, which is the strategy adopted by Meghir et al. (2022), Lagakos, Mobarak, and Waugh (2023), and Barnett-Howell et al. (2026). Instead, we replicate the experimental treatment as literally as possible, and then assess how closely the model reproduces the experimental treatment effects—effectively, non-targeted moments.

We test the model's ability to replicate the treatment effects on migration and economic outcomes observed in each RCT. The model reproduces the migration and expenditure impacts observed in Egger et al. (2022) well. However, it understates the income gains of the Barnett-Howell et al. (2026) information treatments while approximately reproducing their migration impacts. It also substantially understates the migration impacts of the Miner (2026) cash and unemployment insurance treatments. Our paper therefore focuses on interpreting the migration impacts of an unconditional cash transfer in Section 4 and predicting its longer-run dynamics. The model predicts that the migration impact grows beyond the experimental endline but remains modest. In Section 5, we contrast the model's predictions with the experimental results of the other interventions, discuss potential reasons for their divergence, and predict the effects of rolling the experimental interventions out at national level in general equilibrium. The model predicts that treatment effects of UI and information interventions grow over time and are modestly effective at boosting urbanization nationally. Finally, Section 6 assesses the relative importance of each friction

in constraining Kenyan urbanization and incomes, suggesting that understatement of urban incomes is the strongest explicit friction depressing rural-to-urban migration, but most of the observed urban–rural income gap is due to the unexplained residual migration cost. We view these exercises as only indicative given the divergence between model predictions and experimental results for two of the papers we study.

Our paper contributes to the emerging synthesis of development macroeconomics and microeconomic evidence from randomized controlled trials (see Buera, Kaboski, and Townsend (2023) for an overview). RCTs provide credible identification of causal effects in specific contexts, while structural models help interpret these findings, understand mechanisms, quantify general equilibrium effects, and extrapolate findings to broader populations. We follow this tradition by using a rich set of empirical moments to discipline a dynamic general equilibrium model and use it to interpret experimental evidence. Doing so helps us bridge the gap between reduced-form estimates of specific migration barriers[2] and the macroeconomic question of how these barriers jointly determine aggregate productivity and the spatial distribution of labor.[3]

Second, we contribute to the literature on the obstacles to and gains from internal migration by providing a unified framework that helps rationalize findings from multiple experimental interventions. Previous studies have paired a specific experiment with a model tailored to that intervention (Meghir et al. 2022; Lagakos, Mobarak, and Waugh 2023; Miner 2026; Barnett-Howell et al. 2026). In contrast, we build a single environment to simulate the large cash transfer of Egger et al. (2022), small cash transfer and unemployment insurance of Miner (2026), and information provision of Barnett-Howell et al. (2026). The model reproduces the treatment effects of large cash transfers well, but falls short in matching the effects of the latter two interventions quantitatively. This approach allows us to assess the factors that can explain the disparate findings across experiments and provides a platform for comparing the effectiveness of different policy levers in a consistent setting.

Our work is most closely related to Lagakos, Mobarak, and Waugh (2023) and Meghir et al. (2022), who build models to interpret migration subsidy experiments

2. Beyond the experiments we study in Kenya, see, for example, De Janvry et al. (2015), Akram, Chowdhury, and Mobarak (2017), Beam (2016), Beam, McKenzie, and Yang (2016), Cai (2020), Shrestha (2020), and Baseler et al. (2025).

3. For examples in this literature, see Lagakos and Waugh (2013), Gollin, Lagakos, and Waugh (2014), Herrendorf and Schoellman (2018), Bryan and Morten (2019), Morten (2019), Lagakos et al. (2020), Tombe and Zhu (2019), and Morten and Oliveira (2024).

by Bryan, Chowdhury, and Mobarak (2014). Like them, we build a dynamic general equilibrium model of migration. We contribute by incorporating additional frictions—information frictions and unemployment risk—and by seeking to simulate several recent randomized controlled trials at once. Furthermore, our model allows us to assess the general-equilibrium implications of scaling these interventions to the national level and to compare the relative importance of various frictions.

2 Model

In this section, we develop a model of internal migration. The model is designed to capture the key frictions that can impede rural-to-urban migration, as identified by recent experimental evidence.

To be useful for our purposes of matching and interpreting RCT evidence, the model needs to include several features. First, model households must be heterogeneous along several dimensions, to allow for selection into migration and different sampling strategies across experiments. Second, households need to face a borrowing constraint and the means of self-insuring against risk, to capture the liquidity channel emphasized by cash transfer experiments. Third, households need to face urban unemployment risk, to capture the role of unemployment insurance. Fourth, households need to have imperfect beliefs about urban earnings, to capture the effects of information interventions. Finally, the model needs to be rich enough to reproduce several distinct experiments within a single structural environment, so that we can compare different interventions in a consistent setting.

2.1 Households

The model is populated by a continuum of infinitely lived heterogeneous households who make consumption, savings, and migration decisions between a rural (denoted by R) and urban (denoted by U) region. Each period, a household chooses its next period location $l' \in \{R, U\}$ and assets a'.

Preferences and Experience. Households have CRRA preferences over adjusted, or *effective*, consumption, and discount the future at rate $\beta \in (0, 1)$. A household's

period utility is

$$u(c, l, x) = \frac{\tilde{c}^{1-\sigma} - 1}{1 - \sigma}, \qquad \tilde{c} = \frac{c}{1 + m_u \cdot \mathbb{1}(l = U,\, x = 0)},$$

where $\sigma > 0$ is the coefficient of relative risk aversion and $\tilde{c}$ denotes effective consumption. Here $m_u \geq 0$ is a migration disutility parameter, and $x \in \{0, 1\}$ indicates urban experience. Whenever an inexperienced household $(x = 0)$ resides in the urban area $(l = U)$, they perceive their consumption as reduced by a factor of $1 + m_u$. This utility cost persists until the household acquires urban experience, or returns to the rural area. This modeling decision captures a dislike for the urban area which fades with experience there, for example reflecting psychological or cultural adjustment costs.

Urban experience $x \in \{0, 1\}$ evolves stochastically. An inexperienced household $(x = 0)$ residing in the urban area gains experience $(x' = 1)$ with probability π_x each period. Once acquired, experience is permanent as long as the household remains in the city. An experienced household $(x = 1)$ that resides in the rural area loses its experience $(x' = 0)$ with probability π_{-x} per period.[4]

Employment. In the rural sector, all workers are employed in every period. In the urban sector, workers face employment risk (Harris and Todaro 1970). An employed urban worker $(e = 1)$ separates from their job with probability λ_s each period, transitioning to unemployment $(e' = 0)$. An unemployed urban worker $(e = 0)$ finds a job with probability λ_f, transitioning to employment $(e' = 1)$. Rural-to-urban migrants arrive unemployed but can find a job with probability λ_f immediately. Unemployed urban workers receive a flat unemployment benefit b.

Endowments. Households supply one unit of labor inelastically in the region where they reside, but are heterogeneous in their productivity levels z_R and z_U. Each period, each household draws two idiosyncratic productivity shocks ϵ_R and ϵ_U for rural and urban work from a normal distribution with standard deviations $\sigma_{\epsilon,R}$ and $\sigma_{\epsilon,U}$, and correlation ρ_ϵ between the two shocks. Productivities z_R and z_U follow a bivariate first-order autoregressive process in logs, with persistence parameters ρ_R

4. Our modeling of migration experience and the disutility of migration is largely borrowed from Lagakos, Mobarak, and Waugh (2023). It also has a parallel in Meghir et al. (2022), who model a "migration asset" that can be acquired while in the city.

and ρ_U:[5]

$$\log z'_R = \rho_R \log z_R + \epsilon'_R, \qquad \log z'_U = \rho_U \log z_U + \epsilon'_U,$$

where the shocks are jointly normally distributed,

$$\begin{pmatrix} \epsilon'_R \\ \epsilon'_U \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{pmatrix} \sigma^2_{\epsilon,R} & \rho_\epsilon \, \sigma_{\epsilon,R} \, \sigma_{\epsilon,U} \\ \rho_\epsilon \, \sigma_{\epsilon,R} \, \sigma_{\epsilon,U} & \sigma^2_{\epsilon,U} \end{pmatrix} \right).$$

Income and Migration Cost. Labor income depends on location and employment status:

$$y(e, l, z_R, z_U) = \begin{cases} w_R \, z_R & \text{if } l = R, \\ w_U \, z_U & \text{if } l = U \text{ and } e = 1, \\ b & \text{if } l = U \text{ and } e = 0, \end{cases} \tag{1}$$

where w_R and w_U are the rural and urban wages, respectively.

Migration from the rural area to the city requires paying an additive monetary cost $m_c > 0$ in the period before the move, while a move from the city back to the rural area is free. We assume that urban households are myopic about moving costs: while rural households account for m_c when making migration decisions, urban households do not anticipate that they will have to pay m_c if they return to the rural area and then migrate to the city again. This assumption allows m_c to deter rural-to-urban moves while still permitting substantial return migration, which helps the estimated model match the urban population share without absorbing all successful migrants into the city in the stationary equilibrium.

Assets and the Borrowing Constraint. Households hold assets a that earn a return r. Each period, they choose how to allocate their income between consumption c and next period's assets a'. They face a borrowing constraint $a' \geq \underline{a}$, potentially inhibiting households from financing moving costs m_c or supporting themselves while they search for a job in the city.

The household's budget constraint is

$$c + a' + \mathbb{1}(l = R, l' = U) \cdot m_c = (1 + r) \, a + y(e, l, z_R, z_U).$$

5. Note that these productivity processes permit persistent, but not permanent, variation in productivity across households.

Understatement of Urban Income. Each household faces a time-invariant information friction $\gamma \in (0, 1]$. While making decisions from the rural area, the household perceives the urban wage as $\tilde{w}_U = \gamma\, w_U$ rather than the true wage w_U. This misperception reduces the household's perceived return to urban employment and discourages migration, as documented in Baseler (2023) and Barnett-Howell et al. (2026).[6]

In the household's dynamic optimization problem, γ enters as a fixed parameter that scales perceived urban wage income. The household does not anticipate that its beliefs will be revised, capturing the idea that information about urban labor markets is difficult to acquire in rural areas. When a household with $\gamma < 1$ migrates to the city, it discovers the true urban wage and is "surprised" by the higher income. If the household later returns to the rural area, it reverts to its misperceived beliefs (but does not anticipate this reversion while in the city).

Formally, define the household's perceived income as

$$\tilde{y}(e, l, z_R, z_U, \gamma) = \begin{cases} w_R\, z_R & \text{if } l = R, \\ \gamma\, w_U\, z_U & \text{if } l = U \text{ and } e = 1, \\ b & \text{if } l = U \text{ and } e = 0, \end{cases}$$

so that the true income function (1) corresponds to the special case $\gamma = 1$.

2.2 Production

A representative firm in each region operates a Cobb-Douglas technology with constant returns to scale:

$$Y_l = A_l\, K_l^{\alpha}\, L_l^{1-\alpha}, \quad l \in \{R, U\},$$

where A_l is region-specific total factor productivity, K_l is the capital stock, L_l is effective labor measured in efficiency units, and α is the capital share. Firms operate in competitive factor markets, so factor prices equal marginal products:

$$w_l = (1 - \alpha)\, A_l \left(\frac{K_l}{L_l}\right)^{\alpha}, \tag{2}$$

$$r + \delta = \alpha\, A_l \left(\frac{K_l}{L_l}\right)^{\alpha - 1}, \tag{3}$$

6. Our modeling choice is borrowed from Barnett-Howell et al. (2026).

where δ is the depreciation rate of capital. Capital is perfectly mobile across regions, so the net return to capital r is equalized. With constant returns to scale and competitive markets, firm profits are zero.

2.3 Equilibrium

State Space. A household's individual state is the vector $s = (a, z_R, z_U, l, x, e, \gamma)$, comprising assets, rural productivity, urban productivity, current location, urban experience, employment status, and the information friction parameter. The control variables are (a', l'): next-period assets and next-period location.

Recursive Formulation. The household's problem can be written recursively.

A rural household chooses whether to remain in the rural area ($l' = R$) or migrate to the city ($l' = U$):

$$V_R(a, z_R, z_U, x, \gamma) = \max\{V_{R\to R}(a, z_R, z_U, x, \gamma),\ V_{R\to U}(a, z_R, z_U, x, \gamma)\}.$$

The value of remaining in the rural area is

$$V_{R\to R} = \max_{a' \geq \underline{a}} \left\{u(c, R, x) + \beta\, \mathbb{E}\left[V_R(a', z'_R, z'_U, x', \gamma)\right]\right\}$$

subject to $c + a' = (1 + r)\, a + \tilde{y}(1, R, z_R, z_U, \gamma)$, where the expectation is over the next-period productivity endowments (z'_R, z'_U) and urban experience x'. The value of migrating to the urban area is

$$V_{R\to U} = \max_{a' \geq \underline{a}} \left\{u(c, R, x) + \beta\, \mathbb{E}\left[V_U(a', z'_R, z'_U, x', e', \gamma)\right]\right\}$$

subject to $c + a' + m_c = (1 + r)\, a + \tilde{y}(1, R, z_R, z_U, \gamma)$, where the expectation is over the next-period productivity endowments (z'_R, z'_U), urban experience x', and employment status e'.

An urban household with employment status e chooses whether to remain or return to the countryside:

$$V_U(a, z_R, z_U, x, e, \gamma) = \max\{V_{U\to U}(a, z_R, z_U, x, e, \gamma),\ V_{U\to R}(a, z_R, z_U, x, e, \gamma)\}.$$

The value of staying in the urban area is

$$V_{U\to U} = \max_{a'\geq \underline{a}} \left\{ u(c, U, x) + \beta\, \mathbb{E}\big[V_U(a', z'_R, z'_U, x', e', \gamma) \big] \right\}$$

subject to $c + a' = (1+r)\, a + \tilde{y}(e, U, z_R, z_U, \gamma)$. The expectation is over the next-period productivity endowments (z'_R, z'_U), urban experience x', and employment status e'. The value of returning to the rural area is

$$V_{U\to R} = \max_{a'\geq \underline{a}} \left\{ u(c, U, x) + \beta\, \mathbb{E}\big[V_R(a', z'_R, z'_U, x', \gamma) \big] \right\}$$

subject to $c + a' + m_c = (1+r)\, a + \tilde{y}(e, U, z_R, z_U, \gamma)$. A household that moves to the rural area arrives employed, since rural employment is guaranteed.

Information Friction in the Recursive Problem. The information friction parameter γ enters the household's problem through the perceived urban wage. In the value function iteration, a household with $\gamma < 1$ perceives urban wage income as $\gamma\, w_U\, z_U$ rather than the true $w_U\, z_U$, yielding value and policy functions indexed by γ that differ from those computed under accurate information ($\gamma = 1$). Because γ is fixed within the household's problem, the household does not anticipate that its beliefs will change with location.

Crucially, households moving to the urban region never actually experience the understated urban wage. In the simulation of the stationary distribution, the value and policy functions assigned to a household depend on its current location. A rural household with $\gamma < 1$ uses $V(\cdot\,;\gamma)$, the value and policy functions solved under its own γ, and therefore underestimates urban income when evaluating whether to migrate. If such a household does migrate to the city, it is switched to $V(\cdot\,;1)$—the value and policy functions corresponding to accurate information—and is "positively surprised" by the true urban income. Conversely, an urban household with $\gamma < 1$ uses $V(\cdot\,;1)$ while in the city, and therefore does not anticipate reverting to its original misperception upon return. If the household moves back to the rural area, it is switched to $V(\cdot\,;\gamma)$, effectively forgetting what it learned about urban income.[7]

A similar logic is applied to migration-cost beliefs. In the simulation, rural house-

7. This modeling choice captures the idea that accurate information about urban labor markets is hard to acquire in rural areas. The asymmetry between the value function iteration and the simulation is intentional: it allows the model to simulate imperfect information in a computationally cheap way, without having to model the learning process explicitly.

holds use the policies solved under the estimated value of m_c. Urban households, in contrast, use the policies solved under the belief that $m_c = 0$: they "forget" that rural-urban migration has a cost, making them more willing to return to the village in the future, not realizing that the migration cost might "trap" them there for a while.

General Equilibrium. A stationary recursive competitive equilibrium consists of value functions V_R and V_U; household policy functions for consumption $c(\cdot)$, savings $a'(\cdot)$, and location choice $l'(\cdot)$; factor prices $\{w_R, w_U, r\}$; aggregate quantities $\{K_R, K_U, L_R, L_U\}$; and a time-invariant distribution of households over the state space, $\mu(s)$, such that:

1. *Household optimization.* Given factor prices, the value functions and policy functions solve the household's recursive problem defined above.

2. *Firm optimization.* In each region $l \in \{R, U\}$, factor prices equal marginal products as in equations (2)–(3). Perfect capital mobility ensures that the net return r is equalized across regions.

3. *Market clearing*. Labor demand equals labor supply in each region, and total capital demanded by firms equals total capital supplied by households through their savings:

$$L_R = \int_{l=R} z_R \, d\mu, \qquad L_U = \int_{l=U,\, e=1} z_U \, d\mu, \qquad K_R + K_U = \int a \, d\mu.$$

   The aggregate resource constraint is

$$\int c \, d\mu + \delta(K_R + K_U) + M = Y_R + Y_U + B,$$

   where $M = \int m_c \cdot \mathbb{1}(l = R, l' = U) \, d\mu(s)$ is total migration costs paid by all migrating households and $B = \int b \cdot \mathbb{1}(l = U, e = 0) \, d\mu(s)$ is total unemployment benefits. In stationary equilibrium, aggregate net investment is zero. Benefits represent a net resource injection into the economy: they are not financed by any tax on households.

4. *Stationary distribution.* The distribution μ is a fixed point of the transition operator T induced by household policy functions and the exogenous stochastic processes: $\mu = T(\mu)$. The transition operator incorporates the information friction described above: a household with permanent friction parameter γ follows

different policy functions depending on its current location. Define the effective friction as $\gamma^{\text{eff}}(\gamma, l) = \gamma$ if $l = R$ and $\gamma^{\text{eff}}(\gamma, l) = 1$ if $l = U$. The operator T evaluates each household's policy functions at $\gamma^{\text{eff}}(\gamma, l)$, so that rural households use value and policy functions solved under their own γ, while urban households use those solved under accurate information.

3 Estimation

We discipline the model using estimates from the extant literature, Kenyan national accounts data, and Kenyan microdata collected by Barnett-Howell et al. (2026).

3.1 Assigned Parameters

Table 1 summarizes the parameters that are fixed directly rather than estimated through the simulated method of moments. Those parameters are:

Period length (t). We set one model period to six months, approximately the mean duration of a post-treatment migration event in Barnett-Howell et al. (2026). A longer period would compress migration and job-search dynamics too aggressively, while a shorter period would limit the availability of moments in the data.

Discount factor (β). We set the discount factor to $\beta = 0.98$, which is the six-month counterpart to the conventional annual value of 0.96.

Risk aversion (σ). The coefficient of relative risk aversion is fixed at $\sigma = 2$, following Lagakos, Mobarak, and Waugh (2023).

Capital share (α). We set the capital share in both regional production functions to the standard $\alpha = 0.33$.[8]

Depreciation (δ). The depreciation rate is fixed at $\delta = 0.051$, which is the six-month equivalent of a standard annual depreciation rate of 10 percent.

8. This implies a labor share of 0.67, which is significantly higher than the labor share of 0.36 reported by ILOSTAT for Kenya. However, a labor share of 0.36 would make most households in the model derive most of their income from returns on assets, which is inconsistent with the fact that most households in the data primarily rely on labor income.

TABLE 1: Assigned Parameters

Parameter	Value	Source
t	6 months	—
β	0.98	Standard discount factor for a six-month period.
σ	2.00	Lagakos, Mobarak, and Waugh (2023).
α	0.33	Standard.
δ	0.051	Standard annual depreciation rate of 0.10 converted to a six-month period.
A_R	1.00	Rural TFP normalization.
λ_s	0.06	Six-month urban job separation hazard measured for migrants measured by Barnett-Howell et al. (2026).
λ_f	0.42	Six-month urban job finding hazard.
m_u	0.34	Income share inexperienced migrants are willing to forgo in order to return to the rural area, measured by Barnett-Howell et al. (2026).
π_x	0.45	Share of initially rural-preferring migrants who switch to an urban preference within six months after migrating, Barnett-Howell et al. (2026).
π_{-x}	0.68	Share of initially urban-preferring return migrants who switch back to a rural preference within six months, Barnett-Howell et al. (2026).
γ	0.56	Average expected Nairobi income reported by rural households as a share of actual Nairobi income, Barnett-Howell et al. (2026).
b	0.00	No formal unemployment insurance in Kenya.
$\underline{a}$	0.00	No borrowing allowed at baseline.

RURAL PRODUCTIVITY NORMALIZATION (A_R). We normalize rural TFP to $A_R = 1$. Since only relative productivity differences matter for the spatial allocation decisions in the model, this normalization leaves urban productivity A_U free to capture the empirically relevant urban productivity premium.

LABOR MARKET TRANSITIONS (λ_s, λ_f). The urban job separation and job-finding rates are set to $\lambda_s = 0.06$ and $\lambda_f = 0.42$ using six-month transition hazards observed for migrants in the survey conducted by Barnett-Howell et al. (2026).

MIGRATION DISUTILITY (m_u). The utility cost of migration for inexperienced migrants is fixed at $m_u = 0.34$. Empirically, this comes from a question in the sur-

vey conducted by Barnett-Howell et al. (2026) asking urban migrants what rural income would make them willing to return home. Among migrants who prefer the village, the implied compensating differential is 34% of income. We interpret this as a consumption-equivalent disutility from urban life for inexperienced migrants.

Experience transitions (π_x, π_{-x}). We set the per-period probability of gaining urban experience to $\pi_x = 0.45$ and the probability of losing it after returning to the rural area to $\pi_{-x} = 0.68$. In the data, π_x is measured as the share of rural-to-urban migrants that initially prefer the village (i.e. when surveyed in the village, state that they require a higher urban income to be willing to move), but switch to an urban preference after migrating (i.e. when surveyed in the city, state that they require a higher rural income to be willing to return), while π_{-x} is measured as the share of urban-preferring migrants that switch to a rural preference after moving home. Both moments are computed from panel survey data collected by Barnett-Howell et al. (2026). These moments are informative because the model's urban experience state is intended to capture exactly these transitions: households learn to navigate the city, build networks, or otherwise adapt so that the urban consumption penalty disappears. The relatively high values of both parameters suggest that migrants adapt to their new location fairly quickly.

Information friction (γ). We set the urban wage understatement parameter to $\gamma = 0.56$. This matches the finding of Barnett-Howell et al. (2026): on average, rural households underestimate Nairobi income (conditional on demographic characteristics) by 44%.

Unemployment benefit (b). The unemployment benefit is set to $b = 0$ in the baseline, as there is no formal unemployment insurance in Kenya.

Borrowing limit ($\underline{a}$). The borrowing limit is set to $\underline{a} = 0$, prohibiting all borrowing at baseline and forcing households to build up precautionary savings to self-insure against shocks and cover the cost of migration.

3.2 Estimated Parameters

The remaining parameters are estimated using the simulated method of moments. We pick a set of moments that are informative about the underlying structural param-

eters and then search for the parameter values that minimize the distance between the model-generated moments and their empirical counterparts. The parameter-moment mapping described below is not one-to-one, as each parameter influences all moments at once. Still, each parameter-moment pair highlights which moment is particularly sensitive to changes in that parameter and therefore provides the main source of identification for it.

TABLE 2: Estimated Parameters

Parameter	Value	Target moment	Moment value	
			Data	Model
A_U	2.02	URPG	3.28	3.31
ρ_R	0.68	Standard deviation of log rural income	1.57	1.58
ρ_U	0.70	Standard deviation of log urban income	1.46	1.47
$\sigma_{\epsilon,R}$	1.12	Standard deviation of residual rural log income	1.14	1.15
$\sigma_{\epsilon,U}$	1.04	Standard deviation of residual urban log income	1.05	1.06
ρ_ϵ	0.01	Slope of urban vs rural log incomes	0.06	0.07
m_c	3.05[a]	Urban population share	0.32	0.30

[a] Here, m_c is converted into a multiple of mean annual labor income.

URBAN PRODUCTIVITY (A_U). Urban TFP is estimated at $A_U = 2.02$ to match the urban–rural income gap in Kenyan household data.

We measure the urban–rural income gap (URPG) as the exponentiated difference in mean log income,

$$\text{URPG} \equiv \exp(\mathbb{E}[\log y_U] - \mathbb{E}[\log y_R]),$$

where y_U and y_R denote urban and rural household income. For log-normal income distributions, this log-mean URPG measure is identical to the ratio of urban to rural median income. Indeed, the two are close in the data: log-mean URPG is 3.28, compared with a median-income ratio of 3.33. We target log-mean URPG because it is not sensitive to the discretization of the income process in the model.

SHOCK VOLATILITIES ($\sigma_{\epsilon,R}$, $\sigma_{\epsilon,U}$). The shock standard deviations are estimated at $\sigma_{\epsilon,R} = 1.12$ and $\sigma_{\epsilon,U} = 1.04$, using the standard deviations of the residuals in an AR(1) regression of rural and urban log income respectively. These moments isolate

the short-run components of earnings risk, leaving ρ_R and ρ_U to capture the long-run components.

Productivity persistence (ρ_R, ρ_U). The persistence parameters are estimated at $\rho_R = 0.68$ and $\rho_U = 0.70$, targeting the cross-sectional standard deviations of log rural and log urban income, respectively.[9] The identifying logic is the same in both locations: persistent heterogeneity, rather than purely transitory shocks, must account for a large share of the earnings differences that matter for migration selection. Higher persistence makes location-specific earning ability more durable over time, so currently productive rural households expect to remain productive if they stay, while productive urban households expect their comparative advantage in the city to persist after migration.

Cross-location shock correlation (ρ_ϵ). We estimate the correlation between rural and urban productivity shocks at $\rho_\epsilon = 0.01$. The target moment is the slope from regressing urban log income on rural log income for observationally comparable households, where each observation is a cluster formed using a k-means algorithm on survey data from Barnett-Howell et al. (2026). In the model, we use potential urban log income on potential rural log income within a household.

Monetary migration cost (m_c). The monetary migration cost is estimated at $m_c = 3.05$ (expressed as a multiple of mean annual labor income), by matching the share of the Kenyan population living in cities. We treat this parameter as a residual migration friction in the model, capturing any reason Kenyan households may have not to migrate that the explicitly modeled and estimated costs and risks do not account for. Consistent with this role, its estimated value far exceeds the direct monetary cost of moving.

9. We do not target the estimated persistences in panel data because this strategy would ignore permanent productivity differences, resulting in a significantly understated cross-sectional dispersion of productivities and incomes. In our model populated by infinitely-lived households, however, productivity persistence plays a dual role: it is intended both to capture the persistence of transitory shocks and to represent, in a simplified way, long-term productivity differences. For an alternative modeling approach that allows for truly permanent productivity differences, see Trouvain (2026).

4 Reproducing the Egger et al. (2022) Cash Transfer Experiment

This section describes how we map the sampling strategy, intervention design, and outcome measurement of the Egger et al. (2022) cash transfer RCT, compares model-estimated to experimental impacts on migration and expenditure, and uses the model to predict the long-run effects of the intervention. We attempt to reproduce the sampling strategy, treatment design, and outcome measurement of the RCT as closely as the structure of the model allows.

4.1 Reproducing the Experiment

Egger et al. (2022) study the impacts of large, unconditional cash transfers on rural Kenyan households. While not explicitly focused on migration, the intervention potentially relieves liquidity constraints, a migration friction emphasized in the literature and included in our model.

4.1.1 Sampling

Experiment. The experimental sample is drawn from Siaya, a predominantly rural Kenyan county. About one-half of villages were randomized into a treatment group. Within treatment villages, eligibility for the transfer is restricted to poorer households, operationalized by requiring that households have thatched roofs, a condition that about a third of the sample satisfies.

Model. The model represents this sampling choice by taking the poorest third of rural households (ranked by assets) and randomly treating half of them.

4.1.2 Treatment

Experiment. Eligible households receive a large unconditional transfer, roughly equal to 75% of mean annual household expenditure among recipients.

Model. We implement the treatment as a one-time asset transfer equal to 75% of mean annual consumption among recipients.

4.1.3 Measurement

Experiment. The endline survey was conducted 19 months after the initial transfer for the median household. The headline economic outcome is the change in household expenditure relative to control households. The change in the migration rate is also reported.

Model. In the model, outcomes are measured after three six-month periods, or 18 months, which is close to the median experimental horizon. The migration treatment effect is measured as the difference in the share of initially rural households living in the urban region at endline between treatment and control. The consumption treatment effect is measured as the difference in consumption between treated and control households, normalized relative to average control consumption: we map this measure to the experimental definition of household expenditure change.

4.1.4 Results

Table 3 compares the experimental and model treatment effects. In the data, the cash transfer raises expenditure by 11.6% and has no significant migration effect. In the model, the endline consumption effect is 14.3%, while the migration effect is virtually zero. The treatment effects therefore line up well in sign and magnitude. Later in this section, we use our model to investigate the potentially surprising null effect on migration.

Table 3: Cash transfer treatment effects in Egger et al. (2022) and the model

	Migration Rate Change		Consumption Change	
Treatment arm	RCT	Model	RCT	Model
Cash transfer	1.0 pp (1.0)	0.0 pp	11.6%*** (2.4)	14.3%

Note: RCT standard errors are in parentheses; pp denotes percentage points. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

4.2 Long-Run Treatment Effects

Beyond comparing short-run treatment effects to those measured by the RCT, our model allows us to evaluate the predicted medium- and long-run effects of the cash

transfer intervention. The long-run dynamics are theoretically ambiguous: the treatment effect on migration could rise over time if newly unconstrained households are waiting for a favorable urban productivity shock, or it could remain small if the asset transfer was insufficient to cover the necessary costs or if few households are financially constrained from migrating.

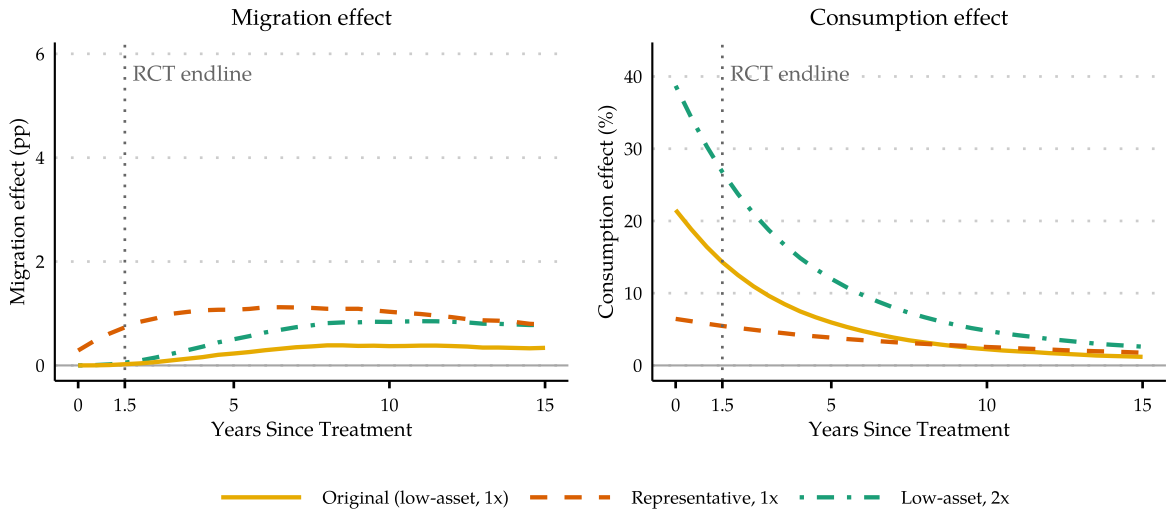


FIGURE 1: Cash transfer emulating Egger et al. (2022): migration and consumption effects over time

Figure 1 shows the migration and consumption effects of the cash transfer intervention emulating Egger et al. (2022). The "Original (low-asset, 1x)" arm reproduces the RCT; the other arms vary sampling or transfer size, as discussed in Section 4.3. For the original arm, the consumption effect decays gradually beyond the endline survey 18 months after treatment as the one-time transfer is spent. The migration effect—indistinguishable from zero in both the model and the experiment at endline—rises slightly over time but remains close to zero. In the model, the treatment impact on migration increases over time as more rural households experience positive urban productivity or negative rural productivity shocks that make migration attractive while retaining enough assets to cover migration costs and unemployment risk. By the 10-year mark, the share of treated households living in the urban area is 0.4 percentage points higher than that of control households; the gap then begins to decay slowly.

TABLE 4: Egger et al. (2022) cash transfer treatment effects in the model: sampling and transfer size

Sampling	Migration Rate Change			Consumption Change		
	LOW-ASSET, 1X	REPRESENTATIVE, 1X	LOW-ASSET, 2X	LOW-ASSET, 1X	REPRESENTATIVE, 1X	LOW-ASSET, 2X
Cash transfer	0.0 pp	0.7 pp	0.1 pp	14.3%	5.5%	26.8%

Note: Low-asset samples randomly treat half of the poorest third of rural households, ranked by assets. Transfer size is the original amount (1x) or twice that amount (2x). Representative sampling randomly assigns households to treatment and control, without restricting eligibility by assets.

4.3 INTERPRETING THE LACK OF MIGRATION RESPONSE

Why does a substantial cash transfer fail to generate a meaningful migration response? Migration is not an inherently immovable outcome: the comparatively light-touch interventions of Miner (2026) and Barnett-Howell et al. (2026) both increased migration. We believe three explanations are possible: few households are financially constrained from migrating, few financially constrained households want to migrate, or the transfer was insufficient to relax the financial constraint for a significant share of households wanting to migrate. To study these potential explanations, we use our model to vary both the size and targeting of the transfer. Specifically, we model the impact of 1) a transfer twice the size of the true transfer targeted in the same way, and 2) a transfer of the original size that is allocated independent of initial assets, in contrast to the true transfer which was targeted toward asset-poor households.

Neither doubling the transfer size nor deploying it to a representative sub-sample of households induces a large migration impact, as shown in Table 4. However, the direction of the changes are informative. Doubling the size of the transfer increases the migration response to 0.8 percentage points after 10 years, after which it stabilizes. The dynamics of the treatment impact on migration look similar to those of the original transfer: there is essentially no change by the endline survey 1.5 years later, followed by a gradual rise which peaks around 10 years. The larger transfer induces a greater migration response because it leaves a greater share of households sufficiently unconstrained that they need only wait for a favorable productivity shock before migrating.

Deploying the original transfer value to a representative sub-sample of households, rather than targeting it toward asset-poor households, also creates a slightly larger migration response of around 1.0 percentage points after 10 years. The impact dynamics are faster compared to the targeted transfers: by the endline the impact

has already reached 0.7 percentage points, compared to 0.0 for the targeted transfer of the same value. Why are asset-richer households more responsive to the transfer? One reason is that productivity shocks in our model are persistent, and households with favorable urban shocks are likely to have recently accumulated more assets. A second reason is that asset-richer households are more likely to be close to the margin of being financially constrained. In our model simulation of targeting the low-asset sample, the average level of pre-transfer assets is only 22% of the estimated migration cost, and ~0.0% of targeted households are able to cover the estimated migration cost immediately after the transfer (building over time as those interested in migration accumulate additional assets on their own). Compare this to the representative sample: the average level of pre-transfer assets is 136% of the estimated migration cost, ~64% of targeted households are unable to afford the migration cost before treatment, and 4.4% of households can newly afford the cost immediately after the transfer. Compared to the ~0.3 pp. immediate impact on migration, this implies that a large share of newly unconstrained households do not want to migrate.

Overall, our model rationalizes the null-to-small impacts of cash transfers on migration by estimating a small share of households that becomes unconstrained from migrating due to the transfer. This is due to the very large migration cost that our model estimates, equal to about three times average annual household labor income. Our model also estimates that, even among newly unconstrained households, a large share will not migrate even over a long period of time because it never experiences a sufficiently favorable productivity shock.

5 Reproducing Miner (2026) and Barnett-Howell et al. (2026) Experiments: UI, Small Cash Transfer, and Information

This section describes how we represent the Miner (2026) and Barnett-Howell et al. (2026) experiments in our model, compares model-estimated to experimental impacts, and uses the model to study the long-run effects of these interventions as well as the aggregate effects of scaling them up.

5.1 Miner (2026): Unemployment Insurance and Cash Transfer

Miner (2026) studies the impacts of smaller unconditional cash transfers, as well as urban unemployment insurance, on rural Kenyan households. The unemployment insurance intervention is aimed at reducing the risk of migrating—captured in our model by urban unemployment risk—by adding an income floor in the city.

5.1.1 Sampling

Experiment. The sample for this paper is drawn from Siaya county, as in Egger et al. (2022). Within sampled villages, the experiment recruits male rural workers aged 18–30. By design, this sampling strategy targets the demographic most likely to migrate: in fact, 65% of participants had prior migration experience. The study covers about 1,300 workers in 111 villages, which corresponds to roughly 11.7 sampled workers per village.

Model. To capture the idea that the recruited young men are more migration-ready than the average rural person, we prioritize sampling experienced households in the model. We set the target sample size to 11.7% of rural households and sample rural households with prior urban experience first, then fill the remaining slots with randomly sampled inexperienced rural households.

5.1.2 Treatment

Experiment. The study includes two treatments. The first is a migration-contingent unemployment-insurance payment tied to being in Nairobi: treated respondents can claim up to 15 days of benefits equal to the average daily urban wage (about 4 USD per day, or 60 USD total). Claiming the benefit required showing up in person at a central office, making it costly for employed migrants to claim. The second treatment is a one-time unconditional cash transfer of 50 USD.

Model. We map the unemployment-insurance arm into the unemployment-benefit parameter b for treated households, with its magnitude matched to 15 days of the average urban wage. This mapping differs from the RCT in that it treats the unemployment benefit as being offered in perpetuity (15 days of average income for every 6-month period that the urban worker spends unemployed). We map the cash arm into a one-time asset transfer equal to $\frac{50}{60}$ of the unemployment insurance amount.

5.1.3 Measurement

Experiment. The main migration outcome is whether the respondent is in Nairobi at the endline survey, conducted about six months after treatment. The main economic outcome is total monthly income of the individual. We normalize the income coefficients and standard errors by the control mean.

Model. We measure outcomes one model period after treatment, corresponding to the six-month experimental horizon. The migration effect is the change in the urban-residence rate from baseline to endline for treated households relative to the same baseline-to-endline change for controls. The income effect is measured with the same difference-in-differences structure, normalized by baseline income.

5.1.4 Results

Unemployment-insurance arm. In the experiment, unemployment insurance raises current Nairobi residence by 8.2 percentage points (standard error = 1.8 pp.). The income effect is statistically insignificant (3.0% increase, standard error = 6.5%). In the model, the same arm increases migration by 2.1 percentage points: the model captures the positive migration response but substantially understates its magnitude. Income in the model rises by 3.3%, similar to the experimental estimate.

Cash-transfer arm. In the experiment, the unconditional cash transfer raises current Nairobi residence by 4.1 percentage points (standard error = 1.8 pp.). The income effect is statistically insignificant (5.5% decrease, standard error = 7.8%). In the model, the cash arm raises migration by 0.1 percentage points, understating the experimental point estimate, and income by 0.4%, within the experimental confidence interval.

Overall, the model substantially underestimates the migration responses relative to the experimental findings. Given the low model-predicted migration response, it is also arguably overestimating income gains from unemployment insurance, because the model-predicted and experimental income impacts are very similar despite much more migration in the experiment than in the model. One potential explanation is selection: unemployment insurance is especially impactful among low-income and risk-averse people (Miner 2026). Because they're highly risk averse, induced migrants are sensitive to unemployment risk and value insurance that mitigates it. Because they have low pre-treatment incomes, a small absolute income gain may be enough to motivate them to move without creating a large change in average income

in the sample. Moreover, risk-averse agents are more likely to migrate for reasons other than income maximization: in particular, to smooth risk. Because agents in our model are uniformly risk averse, it cannot replicate this selection pattern.

In contrast, the model successfully captures the qualitative pattern of a stronger migration response to the unemployment insurance than to the cash transfer. Provision of urban unemployment insurance makes the urban option unambiguously more attractive for all potential migrants, while the cash transfer only makes it more attractive for those who are liquidity-constrained from undertaking an otherwise profitable move.

One remaining puzzle is why the small cash transfer studied in Miner (2026) produced a larger migration response than the larger cash transfer studied in Egger et al. (2022). Our model points to differences in selection of the two experimental samples. As discussed in Section 4.3, the sample in Egger et al. (2022) is restricted to poor households: in the model, this corresponds to sampling from low-asset households, whose average pre-treatment assets are 22% of the migration cost and who are largely unable to afford migration even after the transfer. The sample in Miner (2026) is not restricted to poor households, and furthermore disproportionately includes households with migration experience: in the model, this corresponds to sampling from the entire rural population while preferentially recruiting experienced households. The average pre-treatment assets of the resulting sample are 140% of the migration cost, similar to the counterfactual representative-sample version of the Egger et al. (2022) cash transfer in Table 4: this produces a greater share of sampled households who were unable to pay the migration cost before the transfer and who are just able to afford it after receiving the transfer. Our model thus replicates the difference between the Egger et al. (2022) and Miner (2026) cash transfers qualitatively despite underestimating the magnitude of the cash-induced migration response in Miner (2026).

5.2 Barnett-Howell et al. (2026): Information Provision

Barnett-Howell et al. (2026) study the impacts of information and social network interventions on rural–urban migration in Kenya.

5.2.1 Sampling

Experiment. The experimental sample includes roughly 17,000 rural households, randomly drawn from five predominantly rural counties in Kenya. The sample is

TABLE 5: Unemployment insurance and cash transfer treatment effects in Miner (2026) and the model

	Migration Rate Change		Income Change	
Treatment arm	RCT	MODEL	RCT	MODEL
Unemployment insurance	8.2 pp*** (1.8)	2.1 pp	3.0% (6.5)	3.3%
Cash transfer	4.1 pp** (1.8)	0.1 pp	-5.5% (7.8)	0.4%

Note: RCT standard errors are in parentheses; pp denotes percentage points. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

representative of the rural portions of these counties by design, but not necessarily of rural Kenya overall. Treatment is administered to about 30% of households within sampled villages.

MODEL. To mirror the experimental treated share, the model randomly assigns treatment to 30% of rural households in the panel, with the remaining rural households serving as controls. This captures the treatment intensity in the RCT, while ignoring residual differences between the five study counties and the national rural population, which are modest on several key dimensions.

5.2.2 TREATMENT

EXPERIMENT. The interventions provide information about labor-market conditions in Nairobi, such as typical earnings for various demographic groups. The experiment includes three main delivery arms: information provision to individual households, information provision to individual households coupled with connections to urban residents ("mentors") who can provide additional information, and information provision through group meetings in villages.

Baseline information treatment arm. Enumerators delivered a brochure and read out a script to each household in a one-on-one meeting. As a result, treated households revised their expectations of migration income upward by 4% on average compared to control.

Mentor treatment arm. This arm included the same brochure and script as the baseline information arm, but also paired each household with an experienced ur-

ban mentor offering information over the phone or in person in Nairobi. Expected migration income went up 10% in this arm.

Group treatment arm. In this arm, the same brochure and script were delivered to households in a village-wide group meeting followed by a group discussion. Expected migration income went up 12% in this arm. However, households already experienced in migrating were disproportionately responsive to this treatment arm compared to the baseline and mentor arms, possibly because they were more engaged during and after treatment. Barnett-Howell et al. (2026) use this fact to argue that the group treatment induced adverse selection on gains from migrating.

Model. We take a narrow view of each treatment's impact, mapping the treatments into the model as an increase in rural households' perceived urban earnings.

Baseline information treatment arm. Treatment raises the γ of treated households from $\gamma_{current}$ to $1.04 * \gamma_{current}$, effectively reducing the understatement of urban income and raising expected urban wage by 4%.

Mentor treatment arm. Treatment raises the γ of treated households from $\gamma_{current}$ to $1.10 * \gamma_{current}$, reflecting the 10% increase in expected migration income in this arm.

Group treatment arm. To represent the stronger impact of the group arm on experienced households, we set experienced treated households to $\gamma = 1$ and then infer the change in inexperienced households' γ that yields a 12% average treatment impact on beliefs.

5.2.3 Measurement

Experiment. The main endline outcomes are measured about 16 months after treatment. The paper measures the migration treatment effect as the difference in the share of households that sent at least one migrant to Nairobi between treatment and control. It measures the income treatment effect as the difference in mean household income at endline between treatment and control. The monetary difference is normalized relative to average baseline rural income.

Model. In the model, outcomes are measured after three six-month periods, or 18 months, which is close to the experimental horizon. The migration treatment effect is measured as the difference in the share of initially rural households living in the urban region at endline between treatment and control. The income treatment effect is measured as the difference in the change in mean household income from baseline

to endline between treatment and control, normalized relative to average baseline rural income.

5.2.4 Results

Information treatment arm. In the experiment, the baseline information treatment increases migration by 1.8 percentage points (standard error = 1.0 pp.) and income by 8.9% (standard error = 3.7%). The model generates a similar 0.9 percentage point increase in migration and a smaller 1.2% increase in income.

Mentor treatment arm. In the experiment, the mentor arm raises migration by 2.4 percentage points (standard error = 1.2 pp.) and income by 9.3% (standard error = 4.2%). The model generates a 2.5 percentage point increase in migration and a 3.4% increase in income, matching the migration point estimate closely but understating the income estimate.

Group treatment arm. In the experiment, the group-delivery arm raises migration by 2.2 percentage points (standard error = 1.3 pp.). Its income effect is −2.8% (standard error = 5.1%), statistically insignificant. In the model, the experience-weighted group treatment increases migration by 4.3 percentage points and income by 4.9%. Both model effects exceed the experimental estimates.

Overall, the model replicates the experimental impacts on migration reasonably well: for Information and Mentor the model-predicted impacts are within one standard error of the experimental point estimate, while for Group it is 1.6 standard errors too high. One possible explanation for these divergences is that the treatments affected migration through channels beyond perceived urban earnings: they shared information about employment, prices, and amenities, and the Mentor and Group arms included additional social interactions.

In contrast, the model more substantially under-predicts the income gains from Information and Mentor, and over-predicts that from Group. We believe two explanations can help rationalize these differences. First, our model does not capture general equilibrium effects of migration on rural villages beyond impacts on labor supply, but Barnett-Howell et al. (2026) find evidence consistent with local demand effects à la Egger et al. (2022) generated by income remitted or brought home by migrants. Second, our model incorporates a notion of experience that is transitory and thus rare in equilibrium, meaning that our model is not capable of reproducing the very large degree of selection on the more common measure of experience—

past migration to Nairobi—analyzed in Barnett-Howell et al. (2026). They argue that large differences in engagement with treatment between "experienced" and "inexperienced" households appears largely responsible for driving differences in income gains across treatment arms.

TABLE 6: Information-provision treatment effects in Barnett-Howell et al. (2026) and the model

	Migration Rate Change		Income Change	
Treatment arm	RCT	MODEL	RCT	MODEL
Information	1.8 pp* (1.0)	0.9 pp	8.9%** (3.7)	1.2%
Mentor	2.4 pp** (1.2)	2.5 pp	9.3%** (4.2)	3.4%
Group	2.2 pp* (1.3)	4.3 pp	-2.8% (5.1)	4.9%

Note: RCT standard errors are in parentheses; pp denotes percentage points. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

5.3 LONG-RUN TREATMENT EFFECTS

In this subsection, we use our model to predict the treatment effect dynamics of the Miner (2026) and Barnett-Howell et al. (2026) interventions. Because our model was only partially successful in reproducing these experimental impacts in the short-run, this exercise should be viewed as suggestive.

Figure 2 shows the migration and income effects of the cash transfer and unemployment insurance interventions emulating Miner (2026). By the endline survey 6 months after treatment, less than half of the migration effects of both treatments have been realized. After that, migration effects more than double within five years. Additional rural households migrate as favorable urban or adverse rural productivity shocks make migration attractive now that they can take advantage of the unemployment benefit or cash transfer.

Figure 3 shows the migration and income effects of the information interventions emulating Barnett-Howell et al. (2026). Migration and income gains continue to grow beyond the endline survey 18 months after treatment as more households experience positive urban productivity or negative rural productivity shocks that make

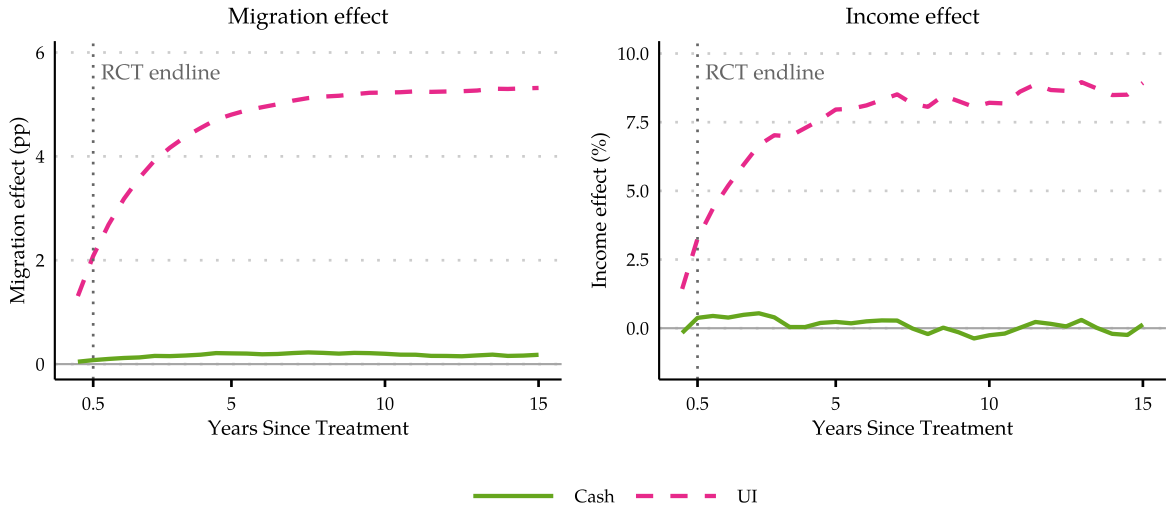


FIGURE 2: Unemployment insurance and cash transfer emulating Miner (2026): migration and income effects over time

migration attractive. A supporting piece of evidence for the gradual ramp-up of treatment effects comes from Barnett-Howell et al. (2026), which finds no significant migration effects at the 8-month midline survey.

5.4 National Rollout

We next use the model to estimate the effects of scaling unemployment insurance and two effective information interventions (individual and mentor) to the national level, accounting for general equilibrium effects of migration through labor and asset markets.[10] To assess the effects of a national rollout in the model, we apply the treatment to all model households and re-solve for the new stationary distribution and general equilibrium. As above, this exercise should be interpreted with caution given the model's ability to only partly reproduce these experimental impacts.

Table 7 shows that all policies are modestly effective: they boost urbanization, lower the urban–rural gap, and increase aggregate consumption. Information interventions raise the urban share by 2.1–4.8 percentage points, while unemployment insurance raises it by 5.2 percentage points. URPG falls modestly from 3.314 in the baseline to 3.242 under individual information provision, 3.178 under mentoring, and

10. Other important spillover channels, which we do not model in this exercise, include congestion or agglomeration effects such as in housing or production markets, multiplier effects of cash in origin economies, and political economy impacts.

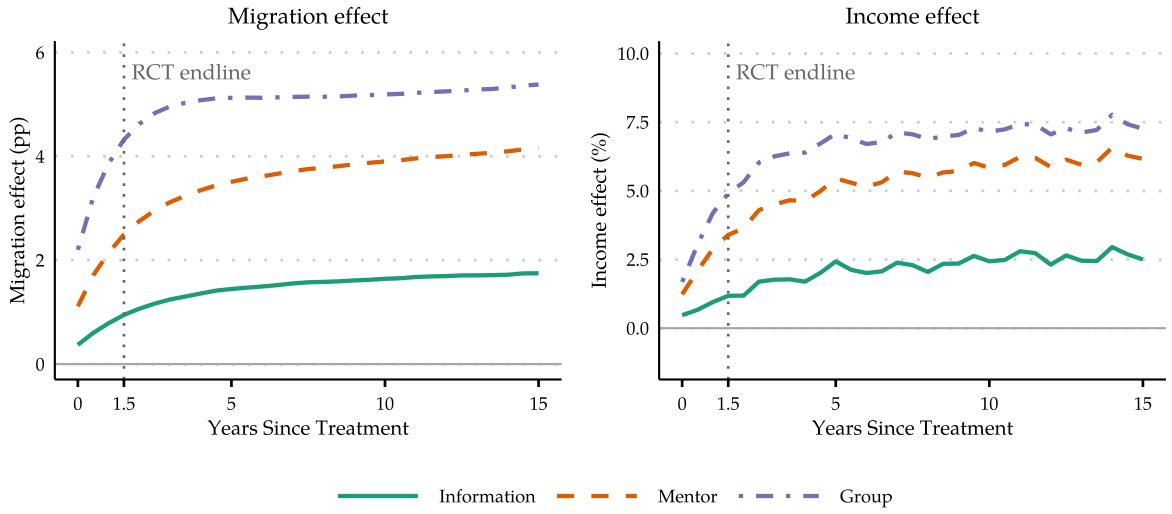


FIGURE 3: Information, mentor, and group interventions emulating Barnett-Howell et al. (2026): migration and income effects over time

TABLE 7: Counterfactual outcomes of rolling out Barnett-Howell et al. (2026) and Miner (2026) treatments nationally

Scenario	Urban Share	URPG	Consumption
Baseline	30.5%	3.314	1.000
Barnett-Howell et al. (2026): Individual	32.6%	3.242	1.018
Barnett-Howell et al. (2026): Mentor	35.3%	3.178	1.040
Miner (2026): Unemployment insurance	35.6%	3.197	1.064

3.197 under unemployment insurance. The information interventions raise aggregate consumption by 1.8–4.0%, while unemployment insurance raises it by 6.4%. Rolling out unemployment insurance nationally would therefore have the largest overall impact among the policies considered. At the same time, it is costlier than the information interventions.

6 Decomposing the Urban–Rural Income Gap

In this section we assess the relative importance of each modeled friction in keeping Kenyan urbanization and incomes depressed. We shut each friction down one at a time and re-solve for the new stationary general equilibrium, thereby isolating the contribution of each friction. Table 8 presents the results. This exercise should be in-

terpreted with caution given the model's ability to only partly reproduce the impacts of interventions targeting information frictions and unemployment risk.

TABLE 8: Results of shutting down one friction at a time

Scenario	Urban Share	URPG	Consumption
Baseline	30.5%	3.314	1.000
No information friction	46.0%	2.869	1.078
No financial friction	36.2%	3.185	1.122
No unemployment risk	33.5%	3.844	1.212
No urban disutility	27.9%	3.502	0.973
No migration cost	77.3%	1.291	1.640

Shutting down the information friction so that rural households do not underestimate potential urban income (setting $\gamma = 1$ for everyone) raises urbanization from 30.5% to 46.0%, lowers URPG from 3.314 to 2.869, and increases aggregate consumption by 7.8%. Of the explicitly modeled frictions, imperfect information appears to be the most significant barrier to migration.

Relaxing the borrowing constraint raises urbanization to 36.2%, lowers URPG to 3.185, and increases aggregate consumption by 12.2%. Greater access to credit allows more households to finance migration and insure against urban employment risk.

Removing the unemployment risk friction (setting $\lambda_s = 0$ and $\lambda_f = 1$) raises urbanization from 30.5% to 33.5% and aggregate consumption by 21.2%. It also raises URPG to 3.844 because eliminating unemployment mechanically increases urban earnings as well as encouraging migration.

Removing the urban disutility (setting $m_u = 0$) reduces the urban share from 30.5% to 27.9%, raises URPG to 3.502, and lowers aggregate consumption by 2.7%. The counterfactual raises rural-to-urban migration by 9% but raises urban-to-rural migration by 23%, lowering urbanization on net. The fear of losing experience therefore retains urban incumbents more strongly than the disutility deters rural households from entering the city.

Finally, eliminating the residual migration cost (setting $m_c = 0$) has the largest effect, raising the urban share to 77.3%, lowering URPG to 1.291 (close to the value of 1 that would indicate no urban–rural income gap), and increasing aggregate consumption by 64.0%. Through the lens of the model, residual migration costs—migration obstacles unexplained by information frictions, inexperience, borrowing constraints, and unemployment risk—are the dominant barrier to migration. After disciplining

the explicit frictions with empirical moments, the model reconciles low urbanization with a large urban earnings premium through the residual migration cost m_c. One interpretation is that travel and relocation costs not captured by migrants' reported compensating differentials are large. Another is that prospective migrants may expect urban experience to accumulate more slowly than the rate estimated among migrants. A large m_c may also absorb permanent productivity differences across households that our strategy of targeting cross-sectional dispersion does not capture.[11]

7 Conclusion

This paper builds a unified quantitative framework to interpret recent experimental evidence on internal migration and to compare the macroeconomic importance of several barriers emphasized in that literature. We find that relaxing liquidity constraints, improving information, and reducing urban employment risk can each increase migration, but with different dynamic and aggregate consequences.

While the model matches the treatment effects of a large cash transfer well, it understates the income gains of information interventions and the migration impacts of smaller cash transfers and unemployment insurance. Heterogeneous risk aversion, local demand spillovers, and richer experience heterogeneity are some mechanisms that the present model lacks that could help explain these shortcomings.

More broadly, our estimates leave an important puzzle unresolved. The frictions we explicitly model—borrowing constraints, unemployment risk, an urban-specific disutility that declines with urban experience, and income misperceptions—appear too small to explain Kenya's low migration rate on their own. To match observed urbanization, the model requires a sizable residual migration cost, suggesting that the main force holding households back is still not fully understood. One possibility is literal travel or relocation costs such as initial job search costs, though these seem unlikely to account for the magnitude of the residual friction by themselves.

In our view, a more plausible interpretation is that prospective migrants overestimate the non-pecuniary difficulty of urban life. Individuals may expect loneliness, disorientation, cultural mismatch, or inability to navigate city labor and housing markets, and therefore assign a large utility cost to moving. Yet once they migrate, many adapt more quickly than anticipated, consistent with the relatively rapid transition

11. See Lagakos and Waugh (2013) for a decomposition of urban–rural gaps into selection based on permanent productivity differences and migration barriers.

from inexperience to experience estimated in the data. Such an interpretation would be consistent with evidence from psychology and behavioral economics that individuals often overestimate the lasting emotional costs of major life changes and underestimate their capacity to adapt once those changes occur (Wilson and Gilbert 2003, 2005). We see attempts to better measure and test this mechanism as a promising direction for future research.

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